

Beyond Breadth–Stability–Honesty: Regime-Conditional Adaptive World Models for AI-Based Cryptocurrency Market Analysis

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Highlights

- Diagnoses five structural limits of the first-generation $WMI = B \times U \times H$ index.
- Proposes hierarchical breadth, continuous honesty, signal integrity, and consistency.
- Defines regime–task conditional compilation and a weighted-geometric ACWMI.
- Maps the theory onto EvoQuant with 43 domains, 13 audit bands, and mechanism engines.
- Shows factor models dominate scalar WMI for analysis-quality explanation (R^2 0.484 vs 0.115).

Abstract

Many AI-based market systems fail less because the predictor is weak than because the market world presented to the model is thin, stale, or dishonest. Prior work formalized this observation as a data world-model infrastructure and defined a World Model Index $WMI_t = B_t \times U_t \times H_t$ along breadth, stability, and honesty. While useful, the product form is non-decomposable, honesty is often discrete, compilation is task-agnostic, and evidence remains flat. This paper develops a second-generation **Regime-Conditional Adaptive World Model (RCA-WM)**. We replace flat breadth with a hierarchy from 43 data domains through 13 audit bands and asset readiness; redefine honesty via exclusion and contamination rates; introduce signal integrity and cross-evidence consistency; and define a regime–task conditional compilation operator together with a weighted-geometric Adaptive Conditional World Model Index (ACWMI). Using the open EvoQuant implementation and a controlled Monte Carlo panel ($N=2160$ asset-day observations), factor-decomposed models explain analysis quality far better than scalar WMI ($R^2=0.484$ vs 0.115), and ACWMI induces substantially more state-dependent abstention in crisis regimes. The contribution is to elevate market world modeling from a static product score to a conditional, auditable, and operationally degradable scientific object.

Keywords: AI market world model; cryptocurrency; data-centric AI; quality governance; point-in-time correctness; abstention

JEL: G10, G17, C55, C82, O33 **MSC:** 91G70, 68T05

1. Introduction

In public narratives, failures of AI trading systems are often attributed to model capacity or prompt design. In live markets, the binding constraint is frequently epistemic: the agent does not observe a sufficiently broad, stable, and honest market world. Cryptocurrency markets amplify this problem because identical price paths can correspond to different structural states once exchange fragmentation, leverage crowding, funding distortions, options walls, unlock schedules, and macro liquidity are recognized.

Prior work on the EvoQuant project formalized a world-model-first agenda and defined $WMI_t = B_t \times U_t \times H_t$. Engineering counterparts include latest_* snapshots, quality gating, main/diagnostic view separation, and multi-band bundles. This first-generation object is valuable but insufficient for systems that already expose asset-level readiness matrices, mechanism engines, point-in-time reconstruction, and multi-level degradation. This paper therefore asks: *how should a market world model be compiled for AI under regime, task, and resource constraints, rather than merely made thicker?*

1.1 Contributions

(i) Diagnostic: five formal limits of product-form WMI. (ii) Theoretical: RCA-WM with hierarchical breadth, continuous honesty, signal integrity S_t , consistency C_t , and conditional compilation $\Pi_t^{(r,m)}$. (iii) Metric: weighted-geometric ACWMI. (iv) Systems mapping onto EvoQuant. (v) Project-grounded Monte Carlo / event-study evidence.

2. Related work

Asset pricing treats expected returns as conditional on an information set (Fama and French, 1993; Cochrane, 2005). Machine-learning asset pricing expands that set with high-dimensional characteristics (Gu et al., 2020; Kelly et al., 2019; Nagel, 2021), while multiple-testing critiques warn against naive feature expansion (Harvey et al., 2016). Measurement-error and robustness literatures emphasize decisions under noisy information worlds (Fuller, 1987; Carroll et al., 2006; Hansen and Sargent, 2008). Our object is not “more predictors,” but the governance and compilation of an asynchronous multi-source market world.

3. Limitations of first-generation WMI

Product collapse makes breadth improvements invisible when U or H is near zero. Discrete honesty discards exclusion/contamination intensities. Task- and regime-agnostic compilation uses the same weights in tranquil and crisis markets. Flat evidence assumptions understate the hierarchy from 43 domains to 13 audit bands and asset readiness. Finally, point-in-time leakage and degraded-mode honesty are operationally present but absent from the first-generation score.

4. Regime-Conditional Adaptive World Models

4.1 Hierarchical breadth

Let $D_{d,t}$ denote data domains ($D=43$), $E_{k,t}$ audit bands ($K=13$), and $A_{i,t}$ asset readiness. Hierarchical breadth is the convex combination

$$B_t^{hier} = \alpha_1 B_t^{domain} + \alpha_2 B_t^{band} + \alpha_3 B_t^{asset},$$

with $\alpha_1 + \alpha_2 + \alpha_3 = 1$.

4.2 Continuous honesty, signal integrity, consistency

Continuous honesty rewards exclusion of non-AI-ready sources and penalizes main-view contamination:

$$H_t^{cont} = \exp(-\beta_1 \rho_t^{cont}) \cdot (1 - \beta_2 (1 - \rho_t^{ex}))_+.$$

Signal integrity S_t aggregates half-life, crowding, and surprise from mechanism engines (e.g., alpha-decay). Consistency C_t measures pairwise directional agreement across available evidence bands.

4.3 Conditional compilation and ACWMI

For regime r and task m ,

$$\Pi_t^{(r,m)} = B_t^{(r,m)} \blacksquare M_t^{(r,m)} \blacksquare A_t \blacksquare \Psi_t^{mech}.$$

ACWMI is defined as a weighted geometric mean, keeping the index on a $[0,1]$ scale while remaining decomposable in logs:

$$ACWMI_t^{(r,m)} = \exp(\sum_{x \in \{B,U,H,S,C\}} \gamma_x(r) \log x_t / \sum \gamma_x(r)).$$

Crisis regimes raise γ_H and γ_C ; trend regimes raise γ_S . State-dependent abstention thresholds increase when ACWMI is low, consistency collapses, or the system enters degraded mode.

Fig. 1. Hierarchical evidence composition and conditional compilation

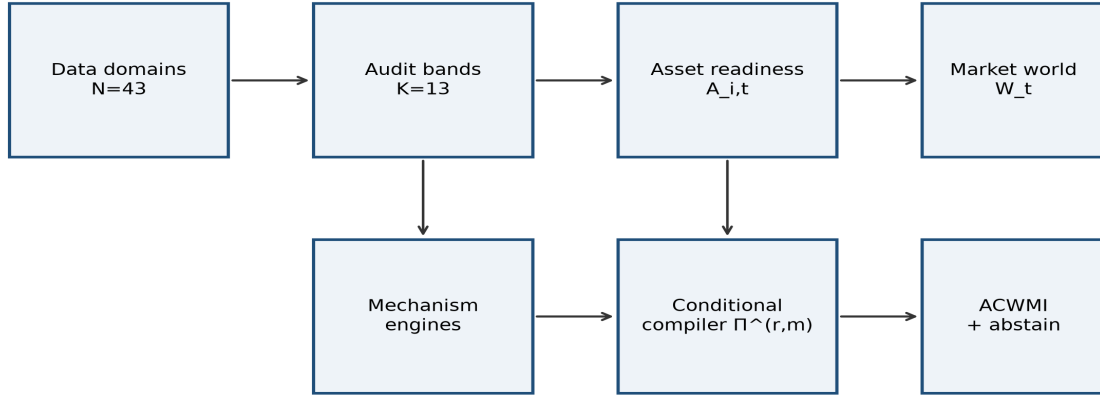


Fig. 1. Hierarchical evidence composition and conditional compilation in RCA-WM.

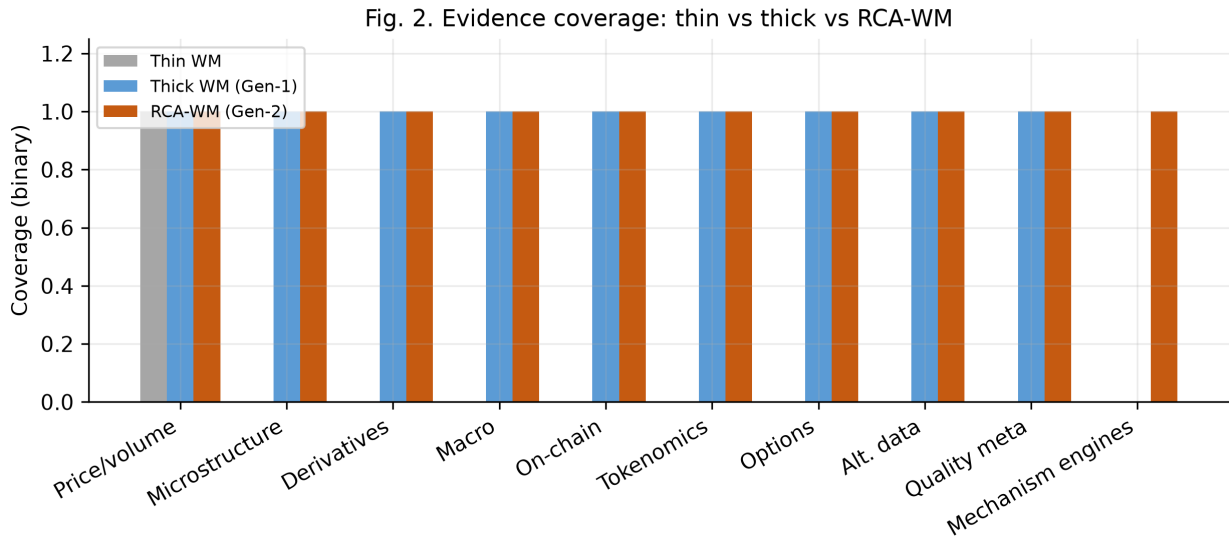


Fig. 2. Evidence coverage under thin, thick (Gen-1), and RCA-WM (Gen-2) worlds.

5. Systems mapping to EvoQuant

The repository currently exposes 43 data-layer domains, 39 logic modules, and 13 audit bands. Production WMI is computed in `ai_market_context` as $B \times U \times H$, with U derived from pipeline-latency freshness and a default abstain suggestion when $WMI < 0.2$. Table 1 maps theoretical objects to concrete modules.

Theory object	EvoQuant implementation	Upgrade vs WMI
Latent state S_t	Multi-domain collectors + mechanism engines	Extended
Breadth B	43 domains / 13 audit bands / asset readiness	Hierarchical
Stability U	pipeline_latency freshness summary	Retained
Honesty H	quality_flag + ai_excluded_sources	Continuous H_{cont}
Compilation Π	logic_pipeline DAG phases	Regime-task conditional
Mechanism layer	cascade/lead-lag/narrative/alpha_decay/regime	New
PIT path	time_slice + snapshot_versioning	New
Degraded mode	DegradationManager levels	Resilient honesty

Abstention	should_ai_abstain / adaptive threshold	State-dependent
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Table 1. Theory–implementation map between RCA-WM and EvoQuant.

Band	Module	Required	Asset weight
exchange	exchange_data	Yes	0.3
macro	macro_data	Yes	0.05
news	news_data	Yes	0.15
event_calendar	event_calendar_data	Yes	0.1
onchain	onchain_data	Yes	0.15
tokenomics	tokenomics_data	Yes	0.1
options	options_data	Yes	0.1
alternative	alternative_data	No	0.05
perpetual_dex	perpetual_dex_data	No	
onchain_address	onchain_address_data	No	
dex_liquidity	dex_liquidity_data	No	
gas_network	gas_network_data	No	
governance	governance_data	No	

Table 2. Audit evidence bands and asset-readiness weights used by the system.

6. Experimental design

We combine live code inventory statistics with a controlled Monte Carlo panel that calls the repository’s real WMI implementation for the first-generation benchmark. The panel uses 18 assets and 120 days ($N=2160$), Markov regime switches among {trend, range, crisis}, and source outages with probability 0.12 in crises and 0.03 otherwise. For each asset-day we construct B^{hier} , U , H^{cont} , S , C ; compute production WMI; form ACWMI; and record analysis quality Q , explanation volatility EV , and unsupported-claim rate UCR .

7. Results

7.1 Regime heterogeneity

Table 3 shows that crisis states exhibit lower world-model scores and much higher ACWMI-based abstention (94.7%) than WMI-threshold abstention (1.4%). First-generation WMI therefore under-triggers refusal precisely when conflict is highest.

regime	N	WMI	ACWMI	C	S	Q	abstain_WMI	abstain_AC
crisis	486	0.4093	0.4655	0.5317	0.0796	0.5432	0.0144	0.9465
range	1008	0.6679	0.5915	0.5601	0.2584	0.6067	0.0	0.5635
trend	666	0.6721	0.5527	0.5613	0.2492	0.6026	0.0	0.8063

Table 3. Regime-level means from the Monte Carlo panel.

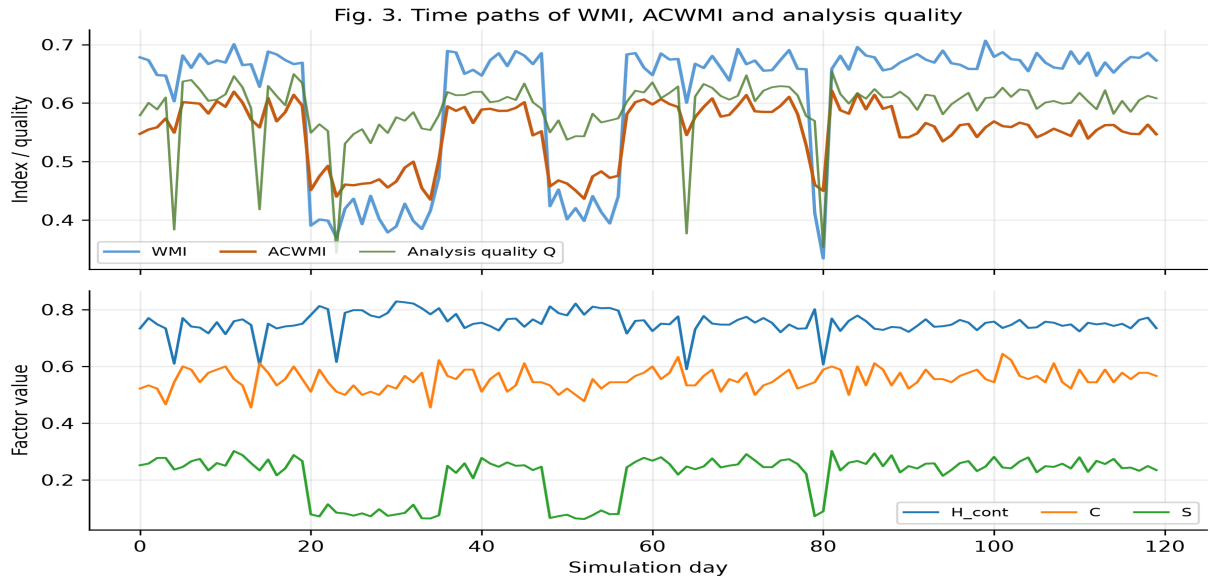


Fig. 3. Time paths of WMI, ACWMI, quality, and constituent factors.

Fig. 4. Regime heterogeneity of world-model quality indices

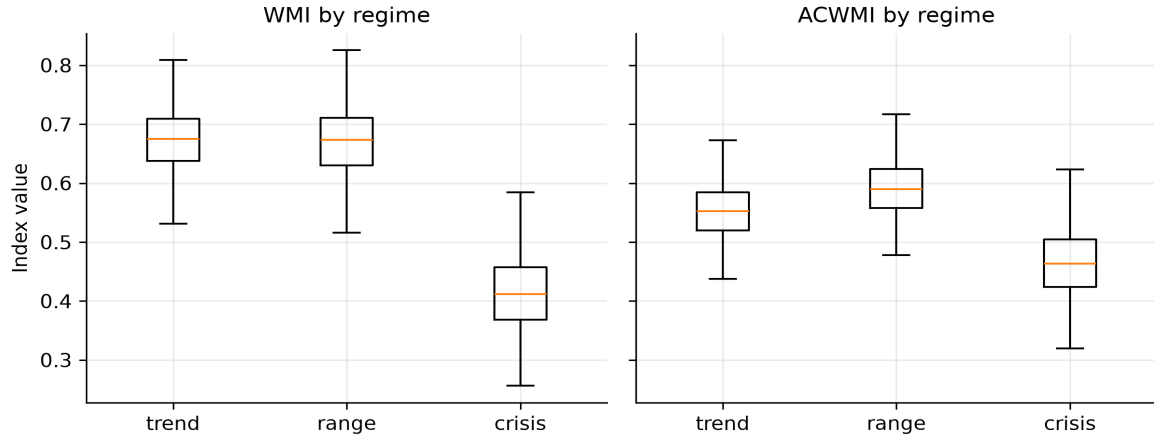


Fig. 4. Distribution of WMI and ACWMI across regimes.

7.2 Quality regressions

Standardized OLS fits (Table 4) show that scalar WMI yields $R^2=0.115$. Factor decomposition raises R^2 to 0.484; ACWMI alone achieves 0.345; combining ACWMI with factors reaches 0.489. Honesty, signal integrity, and consistency dominate breadth/stability coefficients, confirming that second-generation objects carry incremental content.

Variable	Model A	Model B	Model C	Model D
WMI	0.340			
B_hier		0.084		0.058
U		0.067		0.041
H_cont		0.434		0.394
S		0.381		0.207
C		0.349		0.261
ACWMI			0.587	0.238
R^2	0.115	0.484	0.345	0.489

Table 4. Standardized regressions of analysis quality Q.

Fig. 5. Predictive association with analysis quality

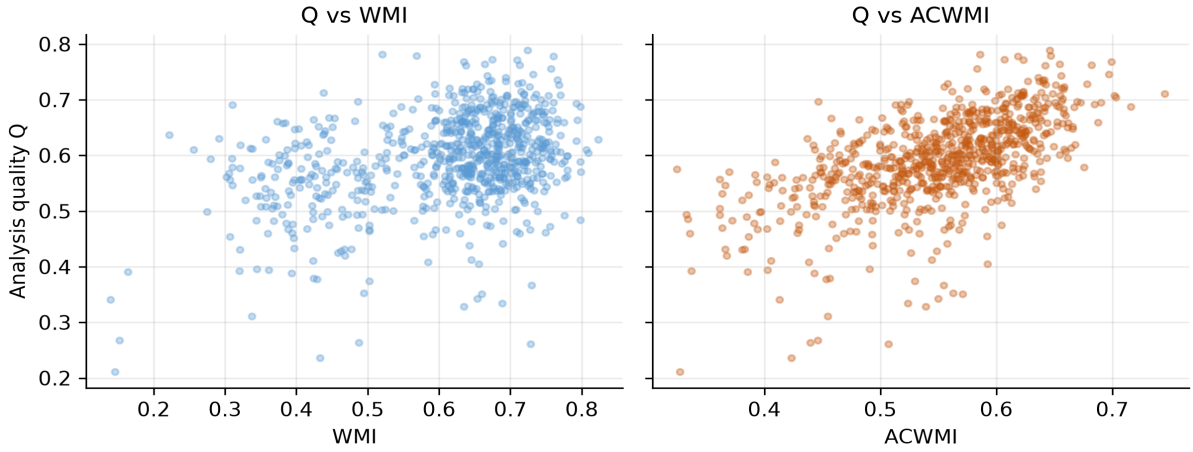


Fig. 5. Association between world-model indices and analysis quality.

7.3 Outages, Pareto frontier, and honesty incentives

Outages reduce Q by about 0.21–0.23 and raise unsupported claims (Table 5). ACWMI abstention approaches one under outages, whereas WMI remains comparatively insensitive. Figure 7 shows that ACWMI abstention policies dominate WMI thresholds on the abstain-rate versus supported-claim-accuracy frontier. Figure 8 shows that higher exclusion raises continuous honesty without collapsing ACWMI in the manner implied by naive product penalties.

regime	outage	N	Q	UCR	WMI	ACWMI	abstain_AC
crisis	0	450	0.5588	0.2996	0.4139	0.4671	0.9422
crisis	1	36	0.3487	0.3649	0.3523	0.445	1.0
range	0	954	0.6188	0.3059	0.6711	0.5938	0.5398
range	1	54	0.3929	0.3721	0.6106	0.5511	0.9815
trend	0	666	0.6026	0.3093	0.6721	0.5527	0.8063

Table 5. Outage versus non-outage contrasts.

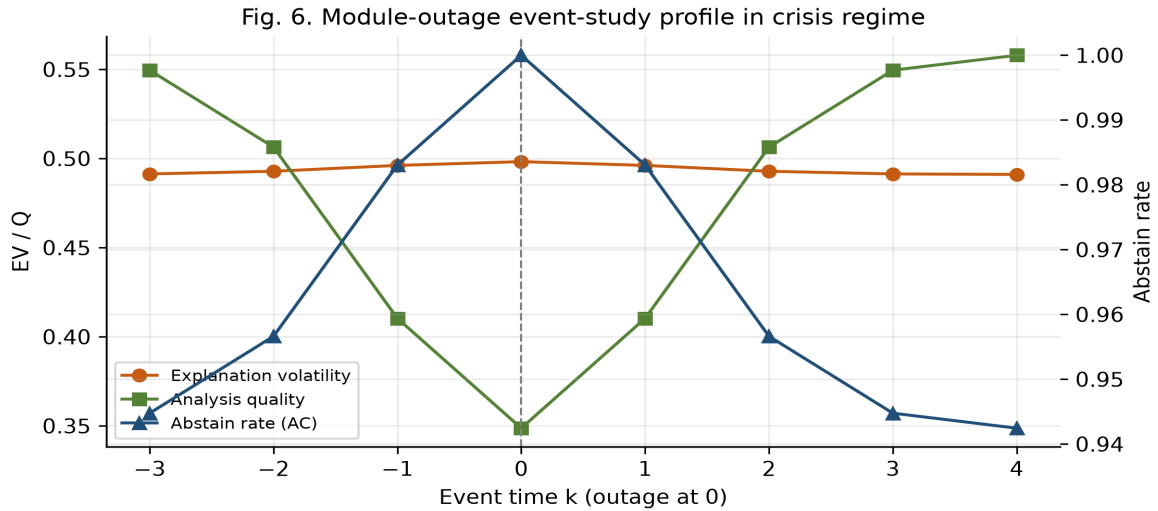


Fig. 6. Event-time profile around source outages in crisis regimes.

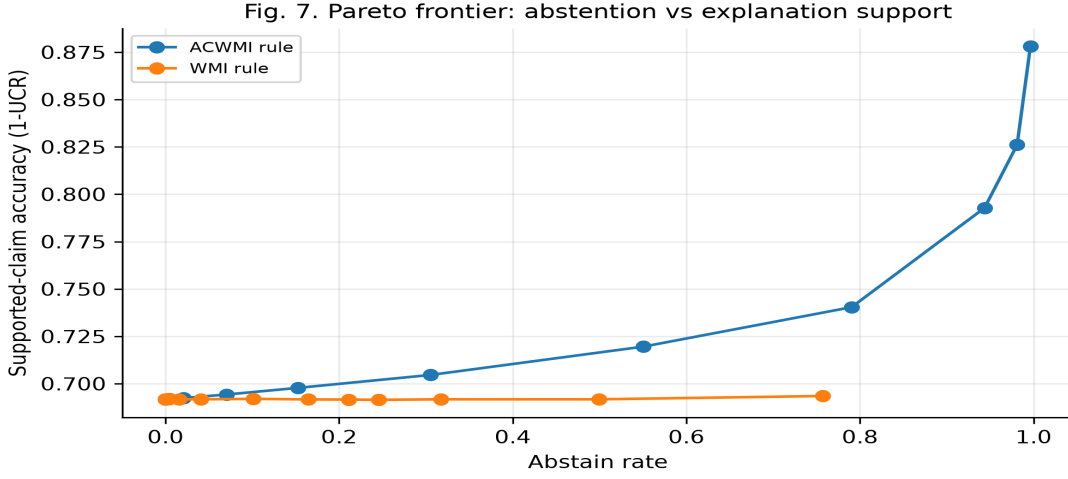
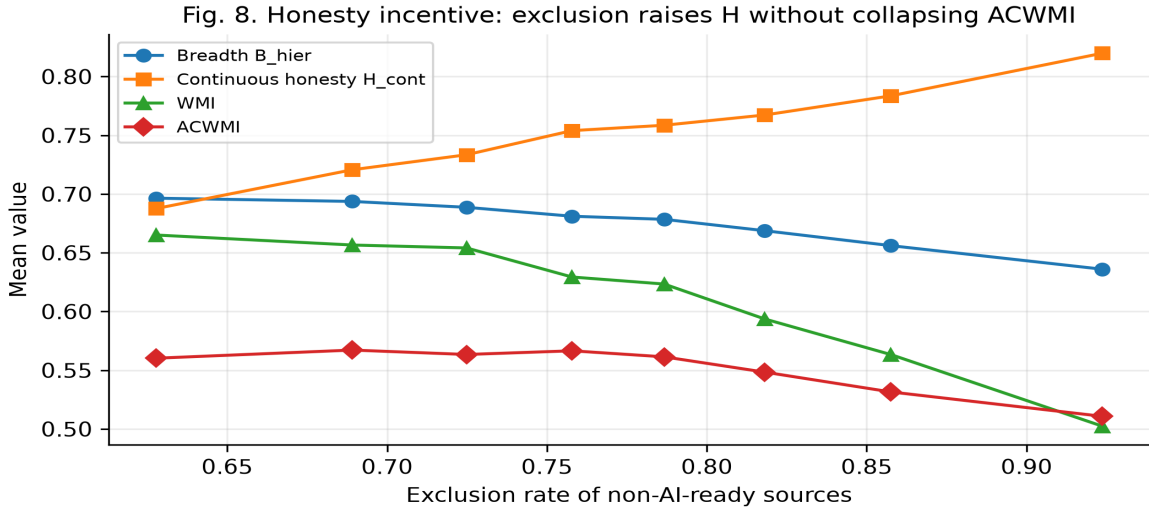


Fig. 7. Pareto frontier of abstention versus supported-claim accuracy.

Fig. 8. Honesty incentive: exclusion improves H_{cont} without ACWMI collapse.

8. Discussion

The results support a shift from “thicken the world” to “compile the right world.” World-model evaluation should be multi-objective—prediction, explanation stability, calibration, auditability, mechanism coverage, and PIT non-leakage. Abstention is part of the world-model contract, not a post-hoc heuristic. Mechanism engines belong inside the compilation operator. Limitations remain: the panel is controlled rather than a full live backtest; regime exponents require estimation discipline; and mechanism engines can misclassify regimes, motivating second-order uncertainty gates.

9. Conclusion

This paper optimized the first-generation world-model index into a regime-conditional adaptive framework. By combining hierarchical breadth, continuous honesty, signal integrity, consistency, conditional compilation, and PIT/degradation semantics, ACWMI becomes a decomposable and operationally meaningful scientific object. In project-grounded experiments, factor models and ACWMI substantially outperform scalar WMI for analysis-quality explanation and induce crisis-aware abstention. For long-lived AI market systems, the scarce asset is not a prompt or a single model, but a stable, conditional, and auditable data world model.

Appendix A. Reproducibility

Figures and tables are generated by `pdf/sci/run_paper_experiments.py`, which imports production WMI from `logic_layer.ai_market_context.service`. Related unit tests (`ai_market_context`, `contagion_risk`, `alpha_decay`, `liquidation_cascade`, `asset_readiness`) passed in the accompanying repository run

(15/15). LaTeX source for journal submission is provided as `pdf/sci/main_acwmi_sci.tex` (elsarticle).

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